Representation of Integers

Computers represent data using binary values (0 and 1). For signed integers, the note describes three methods: **Sign and Magnitude**, **1's Complement**, and **2's Complement**.

1. Sign and Magnitude Representation

In an 8-bit representation, the **MSB (leftmost bit)** represents the sign, and the remaining 7 bits represent the magnitude (value). **0** indicates positive and **1** indicates negative.

For 3:

```
+3 = 00000011
```

For –3, change only the sign bit:

```
–3 = 10000011
```

2. 1's Complement Representation

Positive numbers are stored in their original binary form. For a negative number, obtain its 1's complement by replacing every **0 with 1** and every **1 with 0**.

Example: represent –3 in 8 bits.

```
+3 = 00000011
1's complement: 11111100
Therefore, –3 = 11111100
```

Easy rule: 1's Complement = Flip all bits.

3. 2's Complement Representation

Positive numbers are stored in their original binary form. For a negative number, first find its 1's complement and then add 1.

Example: represent –3 in 8 bits.

```
+3 = 00000011
1's complement = 11111100
Add 1:
11111100
+ 00000001
-----------
11111101
Therefore, –3 = 11111101
```

Easy rule: 2's Complement = Flip all bits + 1.

Comparison: –3 in 8-bit form

Method	Representation of –3	Basic idea
Sign & Magnitude	`10000011`	Change the sign bit
1's Complement	`11111100`	Flip all bits
2's Complement	`11111101`	Flip all bits + 1

Quick Revision

Sign & Magnitude: Change the sign bit.
1's Complement: Flip every bit (0↔1).
2's Complement: Find 1's complement, then add 1.

Remember: The examples above use 8-bit representation, so 3 is written as 00000011.